

A COMPACT CLOSED FORM FOR THE MIDDLE COMPANION ROW OF THE BROWN–ZUDILIN $\zeta(5)$ CONSTRUCTION

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ABSTRACT. Brown and Zudilin’s cellular construction for $\zeta(5)$ produces three rational sequences $Q_n, \hat{P}_n, P_n$, all annihilated by one explicit order-three operator. Only Q_n was given by a closed binomial sum; $\hat{P}_n$ and P_n are defined by the recurrence. We prove that the middle row has the two-term closed form $\hat{P}_n = \sum_{k,l} T(n, k, l) (H_{n+k}^{(3)} - \Psi H_{n+k}^{(2)})$, where T is the cellular summand and Ψ an explicit antisymmetric weight. The proof combines a Barnes–Mellin descent with Zudilin’s partial-fraction lemma, and uses no comparison of $\mathbb{Q}$ -coefficients of zeta values — a point of substance, since the required linear independence is not known. We record what the same methods do *not* give for the top row P_n , in the form of five bounded exclusions. Two companion results are included for contrast: the corresponding $\zeta(3)$ closed form, which is machine-verified; and the $\zeta(2)$ case, where the closed form is known numerically but the method *provably* fails, by an obstruction that is the smallest instance of the one still blocking the top row.

1. INTRODUCTION

Brown and Zudilin [1] construct, from cellular integrals on $\overline{\mathcal{M}}_{0,8}$, a family of linear forms in $1, \zeta(2)\zeta(3), \zeta(5)$ whose coefficients are three rational sequences $Q_n, \hat{P}_n, P_n$. All three satisfy a single order-three recurrence with polynomial coefficients, obtained in [1] by creative telescoping on the integral. The sequences are then *defined* by that recurrence together with initial values; the agreement of two order-three solutions at $n = 0, 1, 2$ is what proves their decomposition.

Of the three, only Q_n is given in [1] by a closed binomial sum. Our object here is the middle row.

Theorem 1.1. *For every $n \geq 0$,*

$$\hat{P}_n = \sum_{k,l=0}^n T(n, k, l) \hat{w}_3(n, k, l), \quad \hat{w}_3 = H_{n+k}^{(3)} - \Psi H_{n+k}^{(2)},$$

with T, Ψ as in Definition 2.1.

Two features of the proof are worth flagging at the outset.

First, it is not a creative-telescoping certificate. The operator L of [1] is *not* a telescoper of $T \cdot \hat{w}_3$; the minimal telescoper of that summand has order seven (§5). The proof instead runs through a Barnes–Mellin descent, where the order-three structure lives.

Second, and less obviously, the proof must avoid comparing coefficients of zeta values. The decomposition of [1] identifies the $\zeta(2)$ -coefficient of an integral with $-4\hat{P}_n$; a Barnes evaluation identifies it with something else; and concluding that the two agree requires $\mathbb{Q}$ -linear independence of $1, \zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3)$, which is not known — indeed the irrationality of $\zeta(5)$ is open. Lemma 4.1 is what removes that step.

Status. Theorem 1.1 is proved. It is *not* machine-verified; §8 states precisely what has been formalised and what has not. The corresponding statement for the top row P_n is open, and §5 records what is excluded.

2. THE OBJECTS

Definition 2.1. Write $H_m^{(r)} = \sum_{j=1}^m j^{-r}$ and $H_m = H_m^{(1)}$. Put

$$T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad Q_n = \sum_{k,l=0}^n T(n, k, l).$$

For $r \geq 1$ set

$$A_r(x) = H_{n+x}^{(r)} - H_x^{(r)}, \quad B_r(x) = H_{n-x}^{(r)} - H_x^{(r)}, \quad C_r = H_{n+k+l}^{(r)} - H_{k+l}^{(r)},$$

and

$$\alpha = A_1(k) - A_1(l), \quad \beta = B_1(k) - B_1(l), \quad \Psi = \frac{1}{2}\alpha + \beta, \quad L_k = -A_1(k) - C_1 - 2B_1(k),$$

with L_l the mirror of L_k under $k \leftrightarrow l$. Let L denote the order-three operator of [1], and let $\widehat{P}$ be the solution of $L\widehat{P} = 0$ with $\widehat{P}_0 = 0$, $\widehat{P}_1 = \frac{101}{4}$, $\widehat{P}_2 = \frac{344923}{96}$.

Note α, β, Ψ are antisymmetric under $k \leftrightarrow l$, while T is symmetric.

3. TWO REDUCTIONS

The first is free, and is used twice more below.

Lemma 3.1. *For any weight w , $\sum_{k,l} T w = \sum_{k,l} T w^{\text{sym}}$, where $w^{\text{sym}}(k, l) = \frac{1}{2}(w(k, l) + w(l, k))$. In particular every $k \leftrightarrow l$ -antisymmetric weight is annihilated by the double sum.*

Proof. $T(n, k, l) = T(n, l, k)$, so relabelling gives $\sum T w = \sum T w^\sigma$ with $w^\sigma(k, l) = w(l, k)$; average. $\square$

Explicitly, writing $X_r = H_{n+k}^{(r)}$ and $Y_r = H_{n+l}^{(r)}$,

$$(1) \quad \widehat{w}_3^{\text{sym}} = \frac{1}{2}(X_3 + Y_3) - \frac{1}{2}\Psi(X_2 - Y_2),$$

since Ψ is antisymmetric.

Lemma 3.2 (the $\zeta(2)$ bridge). *For every $n \geq 0$,*

$$\sum_{k,l=0}^n T \left(H_{k+l}^{(3)} + \frac{1}{4}(L_k + L_l) H_{k+l}^{(2)} \right) = \sum_{k,l=0}^n T \widehat{w}_3^{\text{sym}}.$$

Proof sketch. Both sides arise as the $\zeta(2)$ -coefficient of one Barnes-type double integral. Writing the translated rational kernel as $R(x, y) = P(x)P(y)P(x+y)/(Q(x)^2Q(y)^2)$ with $P(z) = \prod_{r=1}^n (z-r)$ and $Q(z) = \prod_{r=0}^n (z+r)$, the local data at the double pole $y = -l$ is $g_l(x) = c_l P(x)P(x-l)/Q(x)^2$, and $\sum_k B_{kl} = 0$ for each l by vanishing of the residue at infinity. Since $L_k - L_l = -2\Psi$, symmetry reduces the difference of the two sides to $-2 \sum_l \sum_{j=l+1}^n g'_l(j)$; and on $l < j \leq n$ the function g_l has a *double* zero, because $P(x)$ vanishes at $x = j$ (as $1 \leq j \leq n$) and $P(x-l)$ vanishes at $x = j$ (as $1 \leq j-l \leq n$), while $Q(j) \neq 0$ for $j > 0$. Hence every $g'_l(j) = 0$. $\square$

Remark 3.3. The vanishing ranges here are sharp: $g_l(j) = 0$ exactly on $1 \leq j \leq n+l$, $g'_l(j) = 0$ exactly on $l < j \leq n$, and the simple-pole coefficient q_l satisfies $q_l(j) = 0$ exactly on $1 \leq j \leq n$ with a zero of order exactly one.

4. THE DESCENT, AND ITS NORMALISATION

Specialising [1, (I3)] at $k = n + l$ writes the companion integral I_n'' as

$$I_n'' = \sum_l (-1)^l \binom{n+l}{n} \binom{n}{l}^2 J_3(n, n, n, n-l; n, n, n+l),$$

and the rational part of the shifted J_3 is supplied by Zudilin's partial-fraction lemma [2, Lemma 4]: for a suitable one-variable R with $R(t) = \sum_k A_k(t+k)^{-2} + \sum_k B_k(t+k)^{-1}$, the induced quantity is $2A\zeta(3) - B$, where $A = \sum_k A_k$ and

$$B = 2 \sum_k A_k H_{k-h}^{(3)} + \sum_k B_k H_{k-h}^{(2)}.$$

For the specialisation at hand one computes, at $t = -n - 1 - k$,

$$(2) \quad A_{kl} = \binom{n+k}{n} \binom{n+k+l}{n} \binom{n}{k}^2, \quad B_{kl} = A_{kl} L_k.$$

The descent carries a normalisation, and the following lemma is what makes it harmless.

Lemma 4.1 (normalisation). *With the outer factor $(-1)^l \binom{n+l}{n} \binom{n}{l}^2$ of the specialisation,*

$$\binom{n+l}{n} \binom{n}{l}^2 \cdot A_{kl} = T(n, k, l)$$

identically in n, k, l . Consequently the $\zeta(3)$ -coefficient produced by [2, Lemma 4] equals $2Q_n$, whereas that of I_n'' is Q_n ; the descent therefore computes $2I_n''$, and the factor is 2 for every n .

Proof. The two products consist of the same six binomial factors, reordered. The second assertion follows since $Q_n \geq 1$. $\square$

Remark 4.2. This is the step that avoids any independence hypothesis. Both quantities being compared are explicitly computed *rational* coefficients inside a single derivation, not real numbers; no statement about $\mathbb{Q}$ -linear independence of zeta values is used, here or anywhere else in the proof.

Proposition 4.3. *For every $n \geq 0$, $\widehat{P}_n = \sum_{k,l=0}^n T\left(H_{k+l}^{(3)} + \frac{1}{2}L_k H_{k+l}^{(2)}\right)$.*

Proof. Combine the specialisation with (2), [2, Lemma 4] and Lemma 4.1; the outer sign cancels against the $(-1)^l$ of the $\zeta(3)$ -coefficient. $\square$

Proof of Theorem 1.1. By Lemma 3.1 the summand of Proposition 4.3 may be replaced by its symmetrisation, which is $H_{k+l}^{(3)} + \frac{1}{4}(L_k + L_l)H_{k+l}^{(2)}$. Lemma 3.2 identifies its double sum with $\sum T \widehat{w}_3^{\text{sym}}$, and Lemma 3.1 again replaces that by $\sum T \widehat{w}_3$. $\square$

5. WHAT THE SAME METHODS DO NOT GIVE

The top row remains open: we do not know a closed form for P_n . The following exclusions delimit where it is not, and are all obtained by exact computation over stated ranges or by proof, as indicated.

Proposition 5.1. *L is not a telescoper of $T \cdot \widehat{w}_3$. The minimal telescoper of that summand is $A \cdot L$ with A of order four, unique up to scalar, hence of order seven.*

The corresponding certificate is far too large to be useful: its cofactors carry $1.5 \cdot 10^6$ monomials with coefficients of 280 bits and more.

Proposition 5.2. *There is a weight w^* , explicitly a 29-monomial element of the degree-two bare alphabet, with $\sum T w^* = \hat{P}_n$ and such that L is a telescoper of $T \cdot w^*$.*

Telescoper order is thus a property of the summand, not of the sum. This is what makes Proposition 5.1 compatible with the existence of an order-three certificate.

Proposition 5.3. *No identity $T \cdot (w^* - \hat{w}_3) = \Delta_k R_0 + \Delta_l S_0$ with vanishing boundary terms exists. Any operator M with $M \cdot T \cdot (w^* - \hat{w}_3)$ a discrete divergence must annihilate Q ; consequently the minimal such order is at least four.*

Proof. For a monomial M_i maximal in the shift closure the relevant shift matrices are diagonal, so the M_i -component is a scalar equation; multiplying by T and summing over $0 \leq k, l \leq n$ telescopes the right-hand side to boundary terms, giving $d_i Q_n = 0$ where $d = w^* - \hat{w}_3$. Since $Q_n \geq 1$, $d_i = 0$; but d has 29 nonzero maximal components. The second assertion follows from the same computation applied to M , together with the fact that Q has no annihilator of order one or two with polynomial coefficients of degree ≤ 13 . $\square$

Proposition 5.4. *L is not a telescoper of $T \cdot w$ for any weight w with $\sum T w = P_n$ lying in the nine-symbol bare weight-five span of degree at most three.*

This is an exact computation: the admissible space has dimension 449 of 1270, and P_n misses its image under the sum map by 292 of 501 equations. The obstruction is sharp — no single block family excludes, but two pairs do minimally — and the same code returns the positive answer of Proposition 5.2 at weight three when run as a control.

Finally, a negative about the literature route. Zudilin’s higher-weight statement [2, Lemma 19] isolates $\zeta(5)$ for a one-variable very-well-poised R , but does not specialise to the double kernel used here; and [1] explicitly leaves the decomposition of the two-variable universal kernels undone, supplying a descent for I'' only. So the argument of §4 has, at present, no weight-five analogue.

6. THE $\zeta(3)$ ANALOGUE, WHICH IS MACHINE-VERIFIED

For contrast, and because the comparison is instructive, we record the corresponding statement one weight down, where the formalisation *is* complete.

Let $\mathcal{A}(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$, let $a_n = \sum_k \mathcal{A}(n, k)$ be the Apéry numbers, and let b be the solution of Apéry’s recurrence $(n+2)^3 b_{n+2} = P(n+1)b_{n+1} - (n+1)^3 b_n$, $P(n) = 34n^3 + 51n^2 + 27n + 5$, with $b_0 = 0$, $b_1 = 6$.

Theorem 6.1 (machine-verified). *For all $n \geq 0$,*

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}).$$

Theorem 6.2 (machine-verified). *For all $n \geq 0$, $\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (H_{n+k} + H_{n-k} - 2H_k) = 0$.*

Theorem 6.2 has a purely rational proof: with $\varphi(z) = \prod_{i=1}^n (z+i) / \prod_{j=0}^n (z-j)$, the simple poles of φ have residues $(-1)^k \binom{n}{k} \binom{n+k}{k}$ by Lagrange interpolation, and $\text{Res}_{z=m} \varphi^2 = 2\beta_m \sum_{k \neq m} \beta_k / (m-k)$ is antisymmetric in (m, k) , hence sums to zero. No contour argument and no growth estimate is required.

Both are formalised in Lean 4 with no `sorry`, and depend on only the three standard axioms (`propext`, `choice`, and `quotient soundness`). The weight $2H_n^{(3)} - H_k^{(3)}$ appears to be new; we have not located it in the literature.

The scope of Theorem 6.1 deserves the same care as §8 demands of Theorem 1.1: in the formal development b is pinned by the *recurrence* together with $b_0 = 0, b_1 = 6$, not by Apéry’s classical double sum with its nested alternating inner sum. The equivalence of the two is proved on paper by six Zeilberger/Gosper certificates and is not formalised. The situation is thus exactly parallel to the $\zeta(5)$ case, one step further advanced.

7. THE $\zeta(2)$ CASE: WHERE THE METHOD PROVABLY STOPS

The same template applied to the Apéry- $\zeta(2)$ pair produces the closed form but *cannot* prove it, and the failure is sharp enough to locate exactly. We record it because it is the smallest instance of the obstruction that also blocks the top row of §5.

Let $S(n, k) = \binom{n}{k}^2 \binom{n+k}{k}$, $A(n) = \sum_k S(n, k)$ (the sequence 1, 3, 19, 147, 1251, ...), annihilated by $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$, and put

$$(3) \quad B(n) = \sum_k S(n, k) \cdot \frac{1}{5} \left[H_n^{(2)} + H_k(2H_k - H_{n-k} - H_n) \right].$$

Then $L[B] = 0$ holds for $n \leq 25$ by exact computation. We do not know a proof.

Everything in the argument of §6 survives as far as the certificate. With

$$\rho(n, k) = k^2 + k(1 + 6n) - 4 - 15n - 11n^2$$

— degree two, the smallest certificate we have met anywhere — the Wilf–Zeilberger step closes, and the structural boundary identity holds *exactly*: $\rho(n, n+1) = -2(n+1)(2n+1) = -(n+1)^2 \binom{2n+2}{n+1} / \binom{2n}{n}$. The correction decomposes as $A + B_1 H_k + B_2(H_{n-k} + H_n)$ with $B_2 = \rho(n, k+1) / (5(k+1))$. Then it stops.

Proposition 7.1. *The H_{n-k} component of (3) is not Gosper-summable.*

Proof. Write $T(k) = S(n, k)B_2(n, k)$. Then $T(k+1)/T(k) = \frac{a(k)}{b(k)} \cdot \frac{c(k+1)}{c(k)}$ with $a(k) = (n-k)^2(n+k+1)$, $b(k) = (k+1)^2(k+2)$, $c(k) = \rho(n, k+1)$, and $\gcd(a(k), b(k+j)) = 1$ for all $j \geq 0$, the roots being $n, -n-1$ against $-j-1, -j-2$ with n generic. Gosper’s equation is $a(k)x(k+1) - b(k-1)x(k) = c(k)$. Since

$$a(k) - b(k-1) = -n(k^2 + (n+2)k - n(n+1))$$

has degree two with leading coefficient $-n \neq 0$, the degree bound gives $\deg x = 0$. Writing $x = x_0$, the coefficient of k^2 forces $x_0 = -1/n$, and the coefficient of k then reads $n+2$ against the required $3+6n$. Contradiction. $\square$

Independently, the corresponding linear system has no solution for any $\deg x \leq 8$.

Remark 7.2. The diagnosis is structural, not accidental. In the cases where the template does close — the $\zeta(3)$ pair of §6, and likewise Franel’s numbers and s_{10} — the weight is $k \leftrightarrow n-k$ symmetric, the certificate module has rank two, and each component telescopes on its own. Here the weight is genuinely three-lettered in H_k, H_{n-k}, H_n ; the components do not telescope separately, and the weight-two part of any enlarged ansatz is forced to vanish, since $\Delta(Sz) = 0$ implies $z = 0$. Closing this case therefore requires either a different gauge for the weight — the harmonic-monomial fit has a large kernel, and a kernel shift changes the correction — or a genuine $\Pi\Sigma$ -ansatz admitting nested sums rather than rational $\times$ harmonic monomials.

This is the same shape of obstruction as at weight five, where the top period $\zeta(5) + 2\zeta(2)\zeta(3)$ is impure in exactly the sense of carrying two independent letters. The $\zeta(2)$ case is a miniature of it, small enough to be computed in full, and the two available fixes are the same two.

8. FORMAL VERIFICATION OF THE $\zeta(5)$ ROWS: WHAT EXISTS

Theorem 1.1 is a proof on paper. A formalisation in Lean 4 (**ZetaLucas**) is in progress and is *not* complete; we state its position precisely because the distinction is easy to lose.

Verified, with no **sorry** and only the three standard axioms: a reflective polynomial-identity checker; the summand shift calculus; the harmonic letter calculus in the shifted base; the implication “recurrence plus initial values implies the closed form”; and the initial values of the relevant double sums, computed from their definitions.

Not verified: that either double sum satisfies L . Two hypotheses remain quarantined, and the $\zeta(5)$ closed forms are consequently not machine-checked. Two further points are worth recording. First, what the order-three certificate of Proposition 5.2 would give, once complete, is $\hat{P}_n = \sum Tw^\star$ — *not* Theorem 1.1, since by Proposition 5.3 the two representatives cannot be bridged at order zero. Second, during this work a named hypothesis carried in the development was found to be false as stated; every theorem depending on it was **sorry**-free with clean axioms, because **#print axioms** reports what a proof depends on and not whether its hypotheses are inhabited. We mention it because the same trap is available to any development that reports conditional results.

REFERENCES

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